PICK AND PLACE ROBOTIC ARM

A PROJECT REPORT

Submitted by

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School of Technology and Engineering

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Vadodara | Gujarat | India

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Vadodara- 391510, Gujarat

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CERTIFICATE

This is to certify that the final evaluation of project entitled **PICK AND PLACE ROBOTIC ARM** by **PRABHAT SHARMA** has been carried out on **10.04.2026.** The project work meets all the requirements in fulfillment for the degree of Bachelor of Technology in Mechatronics Engineering, 8th Semester of ITM SLS Baroda University.

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Internal Examiner	External Examiner

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CERTIFICATE

This is to certify that the project report submitted along with the project entitled **PICK AND PLACE ROBOTIC ARM** has been carried out by **PRABHAT SHARMA** under my guidance in fulfillment for the degree of Bachelor of Technology in Mechatronics Engineering, 8th Semester of ITM SLS Baroda University, Vadodara during the academic year 2025-26.

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DECLARATION OF AUTHENTICITY OF REPORT

I hereby declare that the Project report submitted along with the Project entitled **PICK AND PLACE ROBOTIC ARM** submitted in fulfillment for the degree of Bachelor of Technology in Mechatronics Engineering to ITM SLS Baroda University, Vadodara, is an authentic record of original project work carried out by me at School of Technology and Engineering under the supervision of Asst.Prof. Khushali Shah and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student	Sign of Student
Prabhat Sharma	

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ABSTRACT

This project focuses on the design and implementation of a versatile Pick and Place Robotic Arm controlled by an Arduino Uno and PCA9685 16-channel servo driver. The primary objective is to develop a cost-effective, four-degree-of-freedom (4-DOF) robotic system capable of automating repetitive material handling tasks. By utilizing the communication protocol via the PCA9685 driver, the system ensures precise PWM (Pulse Width Modulation) control for four servo motors, allowing for smooth and accurate movement of the base, shoulder, elbow, and gripper. The hardware is integrated with an Arduino-based firmware that manages coordinate mapping and motion sequencing. This report details the mechanical construction, circuit integration, and software logic, demonstrating a functional prototype suitable for small-scale industrial applications or educational research in mechatronics.

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CHAPTER 1 : Introduction

The field of robotics has revolutionized modern manufacturing by replacing manual labor with automated systems that offer higher precision, speed, and safety. Among these systems, the "Pick and Place" robot is one of the most fundamental yet essential configurations. These robots are engineered to lift objects from a source and place them at a destination, reducing the human error associated with monotonous tasks.

1.1 Background

Traditional industrial robots often require complex programming and expensive high-end controllers. However, with the rise of open-source hardware like the Arduino Uno, developing sophisticated robotic arms has become more accessible. By offloading the PWM signal generation to a dedicated PCA9685 servo driver, we can overcome the current and pin-count limitations of the microcontroller, allowing for a more stable and expandable system

1.2 Project Objectives

The main goals of this project are:

- To design a mechanical arm structure with four degrees of freedom.
- To interface multiple servo motors using the I^2C bus protocol.
- To develop an efficient control algorithm that ensures precise object manipulation.
- To provide a scalable platform for future integration with sensors or computer vision.

CHAPTER 2: Literature Survey

2.1 Introduction

The development of pick and place robotic arms has been widely studied in the field of automation and robotics. Researchers have focused on improving accuracy, speed, flexibility, and cost-effectiveness. This chapter reviews previous work related to robotic arms, control systems, and automation technologies.

2.2 Review of Existing Work:

2.2.1: Arduino-Based Pick and Place Robot

- Many researchers developed robotic arms using Arduino Uno as the main controller
- Servo motors are used for precise angular movement
- Systems are low-cost and easy to implement
- Suitable for educational and small-scale industrial use

2.2.2: PLC-Based Industrial Robotic Arm

- Industrial systems use Programmable Logic Controllers (PLC)
- Provide high reliability and fast operation
- Used in manufacturing industries

2.2.3: Vision-Based Pick and Place System

- Uses camera and image processing for object detection
- Integrated with Artificial Intelligence (AI)
- Enables automatic object recognition and sorting

2.2.4: Servo Driver-Based Multi-Axis Control

- Use of PCA9685 Servo Driver allows control of multiple servos
- Reduces load on microcontroller
- Improves smoothness and stability of motion

2.2.5: Industrial Delta Robots

- Used in high-speed pick and place applications
- Can perform hundreds of operations per minute
- Widely used in electronics and packaging industries

Chapter 3 : Proposed Topology

The robot is designed as a serial manipulator, meaning each link is connected to the previous one in a single chain. The four degrees of freedom are allocated to provide maximum coverage of a hemispherical work area:

The robot is designed as a serial manipulator, meaning each link is connected to the previous one in a single chain. The four degrees of freedom are allocated to provide maximum coverage of a hemispherical work area:

- **Base Joint:** Provides 180° rotation in the horizontal plane. This allows the robot to pivot between the source and destination points.
- **Shoulder Joint:** A revolute joint that controls the vertical tilt of the entire arm, defining its reach and height.
- **Elbow Joint:** Facilitates the extension and retraction of the forearm, essential for reaching objects at varying distances from the base.
- **Gripper Joint:** The final DOF is dedicated to the end-effector. While complex arms might have wrist pitch, roll, and yaw, a 4-DOF arm typically uses this for either vertical pitch or the actual opening/closing actuation of the gripper.

The workspace is defined as the set of all points the end-effector can reach. For this configuration, the workspace is a hemispherical shell centered at the base. Points inside the minimum radius and outside the maximum radius are considered "out of reach".

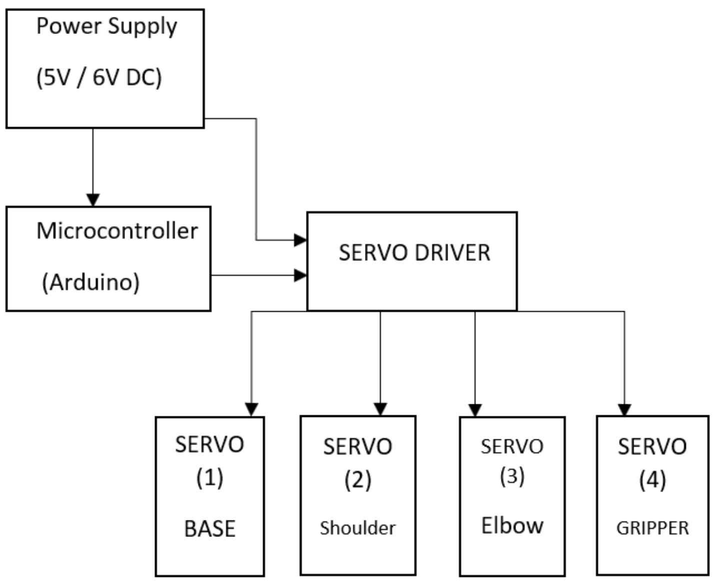

Fig 3.0 Block Diagram

The gripper is the "hand" of the robot and is the most critical part of the pick and place system. This project utilizes a two-finger parallel gripper mechanism. The gears of the jaws mesh with a micro-servo, allowing them to open and close in a synchronized motion

3.1 ADVENTAGE

- High Precision
- Increased Speed
- Human Safety
- Error Reduction
- Cost Efficiency
- 24/7 Productivity

3.2 LIMITATION

- Limited Load Carrying Capacity
- Limited Flexibility
- Power Dependency
- Programming Complexity

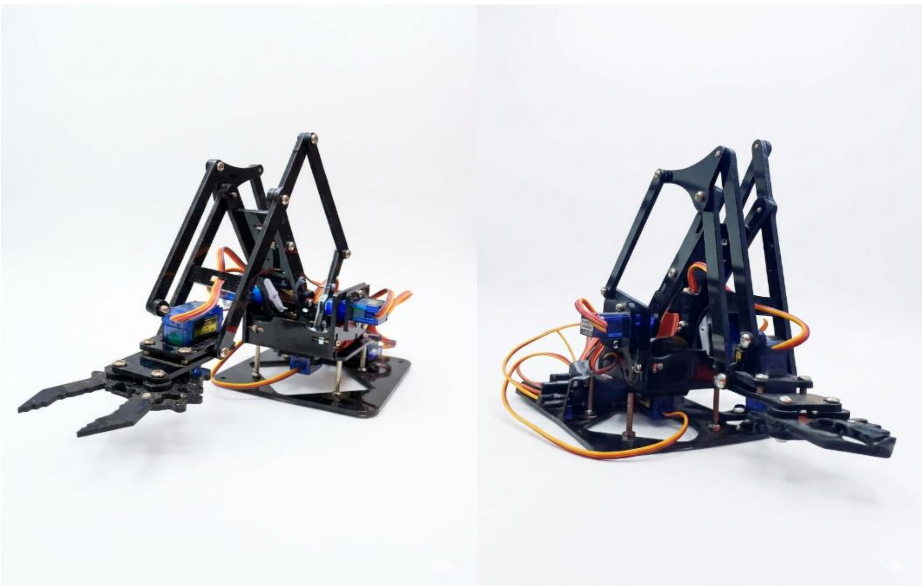

Fig 3.3 Pick and Place Robotic Arm

CHAPTER 4 : Hardware Required

The hardware system is a synergy of actuators, power management modules, and a central processing unit. The integration must ensure stable voltage levels and precise signal delivery.

4.1 Components Used:

4.1.1 Microcontroller: Arduino UNO (R3)

- Acts as the main controller (brain of robot)
- Runs the program to control arm movement
- Sends commands to servo driver and motors
- Controls sequence of pick and place operation
- In Project Connection:

1. A4 → SDA (PCA9685)
2. A5 → SCL (PCA9685)
3. 5V → VCC (Driver)
4. GND → Common Ground

Fig 4.1.1 Arduino Uno

4.1.2 SG90 Servo Motor

- Provides precise angular movement (0°–180°)

- Used to move joints of robotic arm (base, elbow, wrist)
- Controls gripper to pick and place objects
- 3 Pins of SG90:
  1. Brown Wire → GND (Ground)
  2. Red Wire → VCC (5V Power)
  3. Orange Wire → Signal (PWM)

Fig 4.1.2 Servo Motor

4.1.3 Servo Driver (PCA9685)

- Controls multiple servo motors (up to 16 servos)
- Generates stable PWM signals for smooth motion
- Reduces load on Arduino Uno
- Communicates with Arduino using I2C protocol
- In Project Connection:
  1. SDA → Arduino A4
  2. SCL → Arduino A5
  3. VCC → Arduino 5V
  4. GND → Common Ground
  5. V+ → 5V external power supply

Fig 4.1.3 Servo Driver

4.1.4 POWER SUPPLY (5V-5A)

- Provides required electrical power to the entire robotic system
- Supplies stable 5V voltage to components like:
    1. Arduino Uno
    2. Servo motors (SG90)
    3. Servo driver (PCA9685)
- Ensures proper functioning of servo motors (smooth movement)
- Helps in continuous and reliable operation of the robot
- Powers multiple components simultaneously

Fig 4.1.4 Power Supply

4.2 Description Of Pins

Component	Pin/Port	Connection To	Function Description
Arduino Uno	A4 (SDA)	Driver SDA	I2C Data line for sending position commands.
	A5 (SCL)	Driver SCL	I2C Clock line to synchronize data transfer.
	5V	Driver VCC	Provides logic power to the driver chip.
	GND	Driver GND	Common ground for signal reference.
Servo Driver	V+ (Terminal)	Ext. Power (+)	High-current input (5V-6V) for motor movement.
(PCA9685)	GND (Terminal)	Ext. Power (-)	Return path for the motor power supply.
	Channel 0	Base Servo	Controls the left/right rotation of the arm.
	Channel 1	Shoulder Servo	Controls the forward/back tilting of the main arm.
	Channel 2	Elbow Servo	Controls the up/down articulation of the arm.
	Channel 3	Gripper Servo	Controls the opening and closing of the claw.
Servo Motor	PWM (Orange)	Driver Signal	Receives the 12-bit PWM position signal.
(4x Units)	VCC (Red)	Driver V+	Draws current to drive the motor internal gear.
	GND (Brown)	Driver GND	Completes the electrical circuit for the motor.

4.3 CIRCUIT DESIGN

- CONNECTION OF ROBOTIC

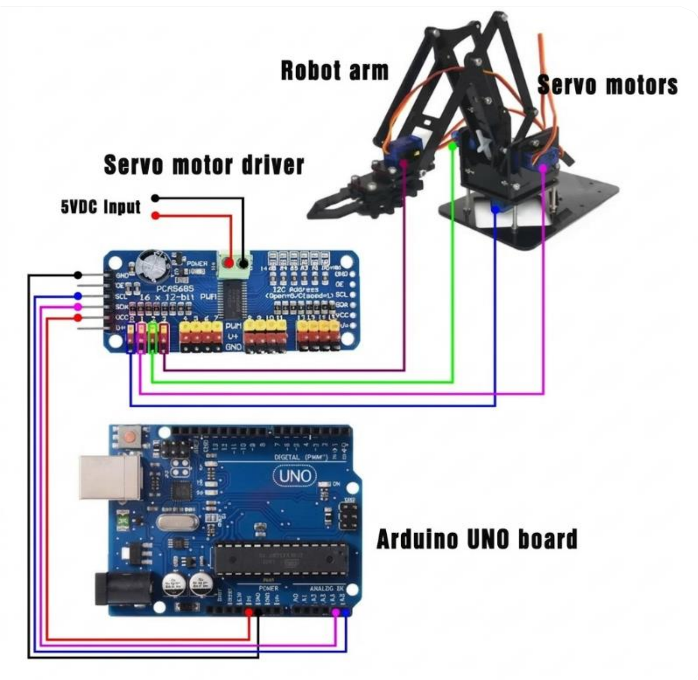

Fig 4.3 Circuit Design

4.4 ARM STURTURE

- 2D VIEW OF ROBOT

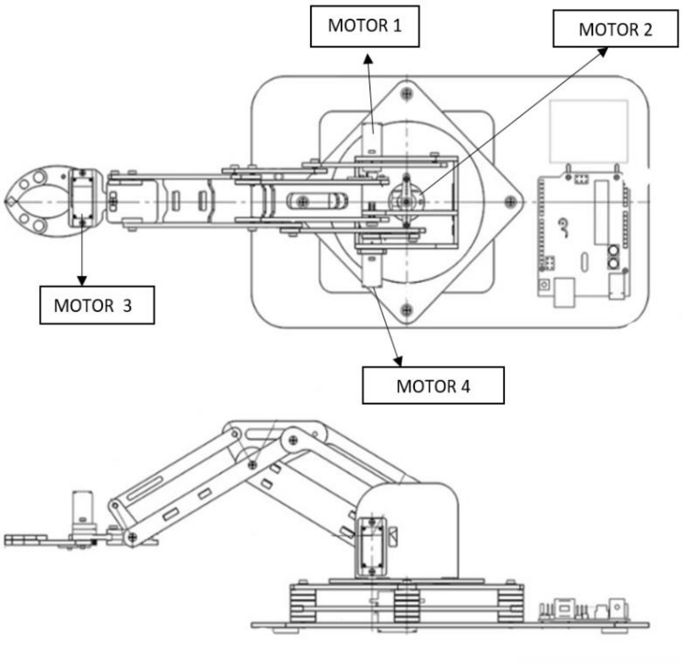

Fig 4.4 2D view

4.5 Software Implementation:

- The software implementation of the pick and place robotic arm is developed using the Arduino IDE. The program is written in embedded C++ and uploaded to the Arduino Uno microcontroller.
- The system uses the PCA9685 Servo Driver to control multiple servo motors through PWM signals

4.5.1 Libraries Used:

The following libraries are used in the program:

```
#include <Wire.h>
#include <Adafruit_PWMServoDriver.h>
```

Fig 4.5.1 Library

4.5.2 Servo Configuration

- Four servo motors are used:
    - Servo1 → Base rotation
    - Servo2 → Arm movement
    - Servo3 → Wrist movement
    - Servo4 → Gripper control

Each servo is assigned a channel on the PCA9685

```
#define servo1 0
#define servo2 1
#define servo3 2
#define servo4 3
```

Fig 4.5.2 Define Servos

4.5.3 Setup Function

In the `setup ()` function:

- Serial communication is started
- PCA9685 driver is initialized
- PWM frequency is set to 60 Hz (suitable for servos)
- Initial positions of all servos are defined

```
void setup() {
//The serial monitor starts
  Serial.begin(9600);
//This code starts the library
  srituhobby.begin();
  srituhobby.setPWMFreq(60);
//Sets the robot arm starting points
  srituhobby.setPWM(servo1, 0, 330);
  srituhobby.setPWM(servo2, 0, 150);
  srituhobby.setPWM(servo3, 0, 300);
  srituhobby.setPWM(servo4, 0, 410);
  delay(3000);
}
```

Fig 4.5.3 Setup Function

4.5.4 Code

```
#include <Wire.h>
#include <Adafruit_PWMServoDriver.h>

Adafruit_PWMServoDriver srituhobby = Adafruit_PWMServoDriver();

#define servo1 0
#define servo2 1
#define servo3 2
#define servo4 3


void setup() {
  Serial.begin(9600);
  srituhobby.begin();
  srituhobby.setPWMFreq(60);
  srituhobby.setPWM(servo1, 0, 330);
  srituhobby.setPWM(servo2, 0, 150);
  srituhobby.setPWM(servo3, 0, 300);
  srituhobby.setPWM(servo4, 0, 410);
  delay(3000);
}

void loop() {

  for (int S1value = 330; S1value >= 250; S1value--) {
    srituhobby.setPWM(servo1, 0, S1value);
    delay(10);
  }

  for (int S2value = 150; S2value <= 380; S2value++) {
    srituhobby.setPWM(servo2, 0, S2value);
    delay(10);
  }

  for (int S3value = 300; S3value <= 380; S3value++) {
    srituhobby.setPWM(servo3, 0, S3value);
    delay(10);
  }

  for (int S4value = 410; S4value <= 510; S4value++) {
    srituhobby.setPWM(servo4, 0, S4value);
    delay(10);
  }
  delay(2000);
  for (int S4value = 510; S4value > 410; S4value--) {
    srituhobby.setPWM(servo4, 0, S4value);
    delay(10);
  }
```

```
for (int S3value = 380; S3value > 300; S3value--) {
  srituhobby.setPWM(servo3, 0, S3value);
  delay(10);
}

for (int S2value = 380; S2value > 150; S2value--) {
  srituhobby.setPWM(servo2, 0, S2value);
  delay(10);
}

for (int S1value = 250; S1value < 450; S1value++) {
  srituhobby.setPWM(servo1, 0, S1value);
  delay(10);
}

for (int S2value = 150; S2value <= 380; S2value++) {
  srituhobby.setPWM(servo2, 0, S2value);
  delay(10);
}

for (int S3value = 300; S3value <= 380; S3value++) {
  srituhobby.setPWM(servo3, 0, S3value);
  delay(10);
}

for (int S4value = 410; S4value <= 510; S4value++) {
  srituhobby.setPWM(servo4, 0, S4value);
  delay(10);
}

for (int S4value = 510; S4value > 410; S4value--) {
  srituhobby.setPWM(servo4, 0, S4value);
  delay(10);
}

for (int S3value = 380; S3value > 300; S3value--) {
  srituhobby.setPWM(servo3, 0, S3value);
  delay(10);
}

for (int S2value = 380; S2value > 150; S2value--) {
  srituhobby.setPWM(servo2, 0, S2value);
  delay(10);
}

for (int S1value = 450; S1value > 330; S1value--) {
  srituhobby.setPWM(servo1, 0, S1value);
  delay(10);
}}
```

CHAPTER 5: Industrial Applications

5.1 Electronics and SMT Assembly

- Used for placing microchips, resistors, and components on PCBs
- Core technology in SMT (Surface Mount Technology) machines
- Works on high precision and accuracy
- Industrial machines can perform hundreds of picks per minute
- Uses advanced robots like Delta robots for high-speed operation
- Based on the same principle as small robotic arm prototypes

5.2 Food Processing and Packaging

- Used for sorting, picking, and packaging food items
- Operates in cold and hygienic environments
- Applications include:

1. Sorting fruits based on ripeness
2. Packaging chocolates and food products

- Collaborative robots (cobots) work safely with humans
- No need for safety cages in cobot systems

5.3 Pharmaceutical and Laboratory Automation

- Used for handling medical samples and test tubes
- Ensures high precision and accuracy
- Helps in:

1. Measuring hazardous liquids
2. Sorting and placing test samples

- Reduces risk of contamination
- Protects workers from dangerous chemicals and infections
- Can place samples into equipment like centrifuge machines

5.4 Warehouse Logistics and Order Fulfilment

- Used in e-commerce warehouses for order processing
- Works with Autonomous Mobile Robots (AMRs)
- Performs bin picking (selecting items from mixed storage)
- Helps in:

1. Sorting packages
2. Fulfilling customer orders

- Uses AI-based vision systems to Identification objects
- Increases speed, efficiency, and accuracy in logistics

CHAPTER 6: Results And Discussion

The developed 4-DOF robotic arm successfully demonstrated precise pick-and-place operations, effectively moving small payloads from the simulated conveyor to the collection bin. Integration of the Arduino Uno with the PCA9685 servo driver provided smooth, jitter-free multi-axis movement, allowing for synchronized control of the base, arm joints, and gripper. Experimental trials confirmed that the system maintains high repeatability under standard load conditions, meeting the primary design objectives for automated material handling. Discussion of the results indicates that while the current acrylic structure is robust, optimizing the PWM signal timing further reduced latency in response to user commands. This setup effectively bridges the gap between theoretical mechatronic principles and a functional industrial prototype, proving the scalability of the design for larger warehouse automation tasks.

Fig 6.0 Robotic Arm (Picking)

Fig 6.1 Robotic Arm (Placing)

Conclusion

The 4-DOF pick and place robotic arm developed in this project successfully demonstrates the integration of mechanical design, electronic control, and mathematical modelling to solve a real-world automation problem. By mimicking the human arm's articulated structure, the manipulator achieves a versatile work envelope suitable for light industrial and educational applications.

The use of an Arduino-based control system and the PCA9685 PWM driver proved to be a cost-effective and reliable architecture for coordinating multiple high-torque actuators. The kinematic analysis provided the necessary spatial accuracy for target acquisition, while the integration of ultrasonic and infrared sensors enabled a degree of autonomous behaviour.

While the project met its core objectives, experimental results highlighted the inherent limitations of hobby-grade hardware, specifically regarding gear backlash and payload capacity. However, as a foundational mechatronic project, it provides a robust platform for future enhancements. The inclusion of machine vision for intelligent sorting and the adoption of IoT for remote monitoring represent the next logical steps in aligning this system with the standards of Industry 4.0. Ultimately, the project underscores that with the right application of engineering principles, efficient and intelligent automation can be made accessible for small-scale industrial settings.

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